

IIoT Sentinel

Industrial IoT Intrusion Detection System (IDS)



CATALOG

- 01** Purpose, Context & Scope
- 02** Core Development Team
- 03** Technical Architecture & Tech Stack
- 04** Functional Modules (A-F)
- 05** Targeted OT Protocols
- 06** Usage Workflow (01-05)
- 07** Conclusion & Key Contributions

Why IIoT Sentinel

- **CONTEXT**

IIoT is central to Industry 4.0; IT/OT convergence increases exposure of critical systems (PLC, SCADA)

PURPOSE

**High-performance IDS
specialized for industrial
networks**

- **SPECIALIZATION**



DoS attack detection on industrial networks



AI-powered forensics of suspicious network frames



Real-time monitoring of industrial protocols
(Modbus, MQTT, S7Comm, OPC-UA)



Map-based visualization of attack flows on OT
network topology

Team & Roles

CORE DEVELOPMENT TEAM (8)



Aya Taftaf
Frontend Developer / Lead Architect



Flahi Fatima-ez-zahraa
Backend Developer / AI Integration



Hajar Bouih
Frontend Developer / UI Design



Flahi Sara
Backend Developer / Network Security



Soukaina Elbaz
Frontend Developer / React Components



Fatima Lachal
Full Stack Developer / Documentation



Aya Boulifa
Frontend Developer / Data Visualization



Aymane Bari
Full Stack Developer / System Architecture

Architecture & Technology Choices



ARCHITECTURE: Modern, component-oriented, real-time; selected for performance, maintainability, and critical monitoring needs



Frontend Core

React + TypeScript



Styling/UI

Tailwind CSS (Cyberpunk/Hacker custom theme)



AI

Google Gemini AI (semantic analysis of complex attack logs)



Build Tool

Vite.js



Visualization

Recharts (real-time network traffic charts)



Icons

Lucide-React

UI Design (Cyber Dashboard)

Visual identity: “Cyberpunk/Hacker” immersion for cybersecurity operations

- ◆ Dark backgrounds (#050505) to reduce fatigue in NOC/SOC environments
- ◆ Green text (#00ff41) referencing classic security terminals
- ◆ Font: JetBrains Mono for log/code readability
- ◆ Scanline + CRT overlay effects for stronger visual identity

```
TERMINAL_V1.0                                STATUS: ACTIVE
> INITIALIZING IIOT_SENTINEL...
> LOADING CORE_ARCH...
> ALERT ANOMALY DETECTED ON PORT 502
> SOURCE: 192.168.1.45
> PROTOCOL: MODBUS/TCP
-
```

Functional Module A — Dashboard

- **ROLE**

Main NOC/SOC command center
aggregating critical metrics

CRITICAL THRESHOLD

1,000 PPS

Automatic switch to Emergency Alert
visual when traffic exceeds this limit.

- **KEY CAPABILITIES**



Real-time AreaChart of packets per second (PPS) with
simulated DoS spikes



Incoming bandwidth, target PLC CPU load, active threat
status



Continuous raw log stream (e.g., SRC:
192.168.1.x | LEN: 1500 | ANOMALY_DETECTED)



Automatic switch when traffic exceeds 1,000 PPS
(Emergency Alert visual)

Functional Module B — AI Forensics

- **ROLE**

Understand and dissect complex attacks using generative AI

- **OPERATOR FLOW**

1 Paste a raw log or suspicious network frame



2 Send to Google Gemini API with an industrial cybersecurity expert prompt

- **OUTPUT: AUTOMATICALLY GENERATED STRUCTURED JSON REPORT**

"Summary":

Simplified explanation for quick SOC reading

"Vectors":

Identified attack vectors (e.g., Buffer Overflow, SYN Flood, UDP Amplification)

"Remediation":

Immediate corrective actions (segmentation, PLC hardening, rate limiting)

Functional Module C — System Terminal

- **ROLE**

Integrated command-line interface for technical operators

- **AVAILABLE COMMANDS**

[Help](#)

List all commands

[Scan](#)

Port scan simulation (1–1024) and detect active services (e.g., Modbus port 502)

[Status](#)

Show IDS kernel status, system memory, firewall status

[Alert](#)

Show latest critical alert and applied mitigation measures

[clear](#)

Clear terminal



```
> sentinel --status
IDS Kernel: ACTIVE
Firewall: REINFORCED

> scan --port 502
MODBUS detected on 192.168.1.10
```

Functional Module D — Attack Logs

- **ROLE**

Immutable registry for post-incident investigation

- **EACH ENTRY CONTAINS**

Timestamp	Precise detection timestamp
Source IP	Source IP address
Vector Type	Vector type: TCP SYN Flood, UDP Amplification, ICMP Flood, HTTP GET Flood
Intensity	Intensity in PPS
Severity	Severity color coding: CRITICAL (red), HIGH (orange), MEDIUM (yellow)

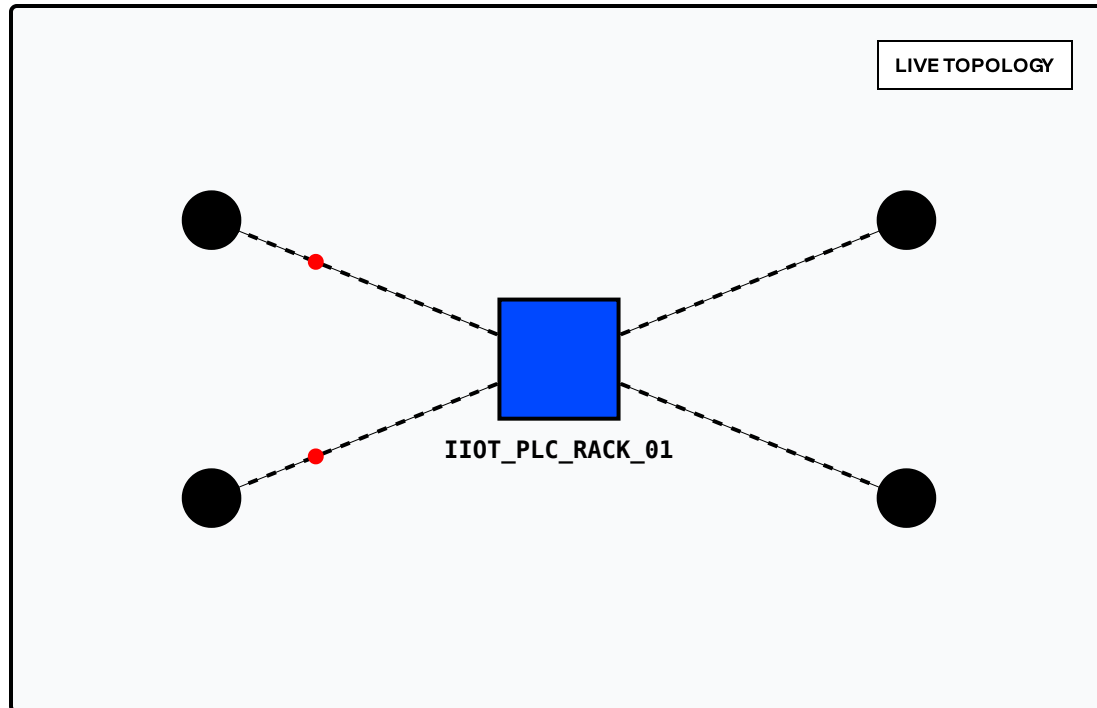
SEVERITY LEVELS

-  **CRITICAL**
-  **HIGH**
-  **MEDIUM**

Functional Module E — Network Map

- **ROLE**

Observe industrial topology and visualize how malicious packets converge to a target



- **WHAT IT ENABLES**



- Identify entry vectors**

Pinpoint exact network entry points used by attackers



- Support segmentation decisions**

Data-driven isolation of compromised network segments

EXAMPLE TARGET

rack PLC IIOT_PLC_RACK_01

Functional Module F — Project Info

- **ROLE**

Embedded documentation module inside the interface

- **ACCESS**

available directly from the UI without leaving the application



- **INCLUDES**



Technical context



User guide



Development team presentation

Industrial Protocol Coverage

🎯 **Focus:** OT protocols (vs. classic IDS focus on HTTP/FTP/SSH)

Modbus/TCP

port 502

PLC communications; common in manufacturing & energy

MQTT

port 1883 / 8883

lightweight pub/sub for industrial sensors & actuators

S7Comm

port 102

Siemens proprietary protocol for Step7/TIA Portal PLCs

OPC-UA

port 4840

interoperability standard for secure equipment communication

Incident Handling Workflow (Part 1)

01

Detection:

- Dashboard turns red; AreaChart shows traffic spike to 4,500 PPS (threshold 1,000 PPS)
- Emergency Alert mode activates automatically

02

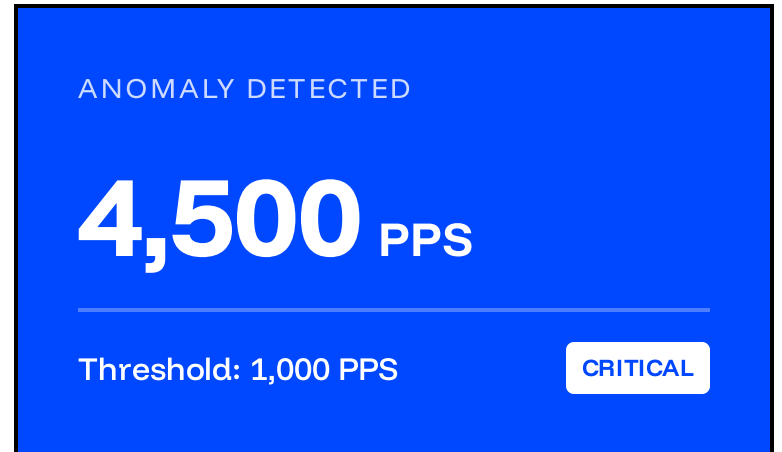
Identification:

- Analyst opens Attack Logs
- Identifies source IP and attack vector (e.g., TCP SYN Flood) with CRITICAL severity

03

Investigation:

- Analyst copies raw frame from Dashboard Live Frame Stream
- Pastes into AI Forensics for deeper analysis



Active Alert Mode

System automatically triggered Emergency Alert protocols upon exceeding the safety threshold.

Incident Handling Workflow (Part 2)

04

AI Analysis:

- Gemini confirms SYN Flood targeting Modbus port 502 on the PLC
- Generates JSON report (Summary, Vectors, Remediation)

GENERATED JSON REPORT



```
"Summary": "SYN Flood detected..."  
"Vectors": ["TCP SYN Flood", "Port 502"]  
"Remediation": "Apply rate limiting..."
```

05

Remediation:

Apply recommendations via System Terminal:

 Activate rate limiting

 Isolate attacked network segment

 Harden industrial firewall rules

Conclusion

IIoT Sentinel is an advanced PoC coupling modern AI (LLMs) with classic network monitoring for industrial cybersecurity

- **KEY CONTRIBUTIONS:**



Reduced MTTR through automated AI forensics



Immersive, ergonomic interface for SOC critical environments



Specialized OT protocol coverage (Modbus, MQTT, S7Comm, OPC-UA)



Modular architecture supporting future extensions (new modules, SIEM connectors)

◆ **Application domains:** critical infrastructures (factories, dams, power plants, water treatment)

Thank You

IIoT Sentinel — Technical Report (2026)